Lab 6: Causal Mediation Analysis

Causal Machine Learning

Stijn Vansteelandt and Federico Bertoia, Ghent University

JOBS II is a randomized field experiment that investigates the efficacy of a job training intervention on unemployed workers. The program is designed not only to increase reemployment among the unemployed but also to enhance the mental health of the job seekers. In the experiment, 1 801 unemployed workers received a prescreening questionnaire and were then randomly assigned to treatment and control groups. Those in the treatment group participated in job skills workshops in which participants learned job search skills and coping strategies for dealing with setbacks in the job search process. Those in the control condition received a booklet describing job search tips. In follow-up interviews, two key outcome variables were measured: a continuous measure of depressive symptoms based on the Hopkins Symptom Checklist and a binary variable, representing whether the respondent had become employed.

Researchers who originally analyzed this experiment hypothesized that workshop attendance leads to better mental health and employment outcomes by enhancing participants' confidence in their ability to search for a job. In the JOBS II data, a continuous measure of job search self-efficacy represents this key mediating variable. The data also include baseline covariates measured before administering the treatment. The most important of these is the pretreatment level of depression, which is measured with the same methods as the continuous outcome variable. There are also several other covariates that are included in our analysis to strengthen the validity of the key identifying assumption of causal mediation analysis. They include measures of education, income, race, marital status, age, sex, previous occupation, and the level of economic hardship.

1. Construct a causal diagram that reflects the design of this experiment and clearly formalizes the mediational hypothesis of interest and possibly also includes variables that need to be accounted for in the final mediation analysis.
2. Use debiased machine learning to estimate the natural direct and indirect effects (via job search self-efficacy) of treatment on the chance of finding work.
   (a) Report the effects along with 95% confidence intervals.
   (b) Interpret the obtained direct and indirect effects in detail.
   (c) What causal assumptions are needed to justify this interpretation? Which of these assumptions appears plausible or is guaranteed to hold? If you cannot judge their plausibility, then state what question you would ask a subject-matter expert to be able to judge their plausibility?
3. Use debiased machine learning to estimate the natural direct and indirect effects (via job search self-efficacy and finding work) of treatment on the depression score.
   (a) Report the effects along with 95% confidence intervals.
   (b) Interpret the obtained direct and indirect effects in detail.

(c) What causal assumptions are needed to justify this interpretation? Which of these assumptions appears plausible or is guaranteed to hold? If you cannot judge their plausibility, then state what question you would ask a subject-matter expert to be able to judge their plausibility?

4. Use debiased machine learning to estimate the natural direct and indirect effects (via job search self-efficacy) of treatment on the depression score.

   (a) Report the effects along with 95% confidence intervals and interpret the obtained direct and indirect effects in detail.

   (b) Take the difference between the indirect effects from question 3 and 4(a). How can we interpret this? Hint: check the causal diagram to understand which pathways correspond to the indirect effects from the previous two analyses.

5. Suppose you were given the efficient influence curves of the indirect effect estimands from the previous analysis 4(b). Then how would you calculate a 95% confidence interval for the difference estimate obtained in the previous question? Hint: the efficient influence curve of the difference between the estimands equals the difference of the respective efficient influence curves.